Determinants

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

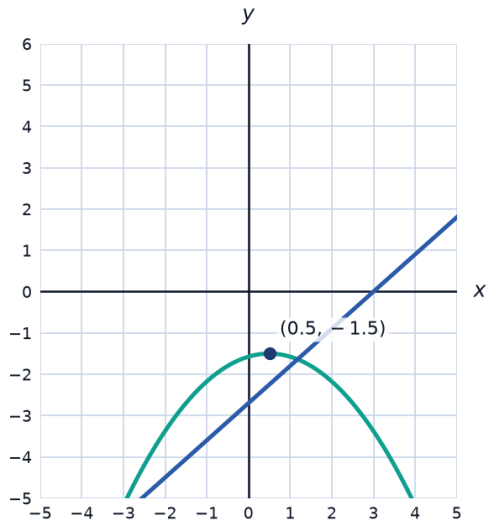
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: 2x2 determinants**

Find the determinant of the matrix with rows $\langle 4, 3\rangle$, $\langle 2, 5\rangle$.

2 **[Foundational]** • **Target: 3x3 determinants via cofactor expansion**

Find the determinant of the matrix with rows $\langle 1, 2, 3\rangle$, $\langle 0, 1, 4\rangle$, $\langle 5, 6, 0\rangle$ by cofactor expansion along the first row.



3 **[Foundational]** • **Target: 3x3 determinants via cofactor expansion**

Recompute the determinant of the matrix with rows $\langle 1, 2, 3\rangle$, $\langle 0, 1, 4\rangle$, $\langle 5, 6, 0\rangle$ by expanding along the SECOND row instead, to confirm the answer matches.

4 **[Foundational]** • **Target: The determinant product rule**

For A with rows $\langle 1, 2\rangle$, $\langle 3, 4\rangle$ and B with rows $\langle 2, 1\rangle$, $\langle 0, 3\rangle$, verify $\det(AB) = \det(A)\det(B)$.

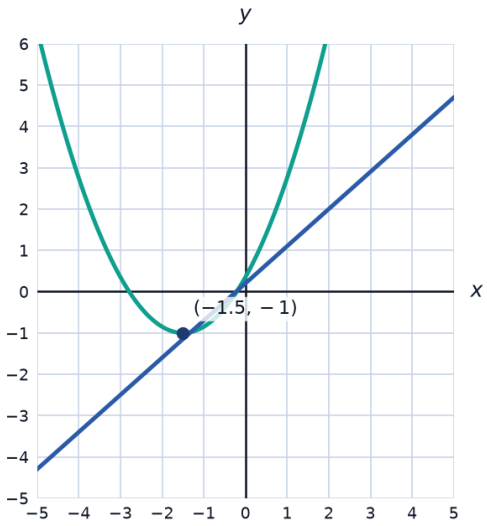
5 **[Foundational]** • **Target: The determinant product rule**

Using the product rule $\det(AB) = \det(A)\det(B)$, explain why $\det(A^{-1}) = \frac{1}{\det A}$ for an invertible matrix A.

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6 [Foundational] • Target: Determinants and scalar multiples / row operations

If A is a 2×2 matrix with $\det A = 5$, find $\det(2A)$ (scaling EVERY entry of A by 2).



7 [Foundational] • Target: Determinants and scalar multiples / row operations

If A is a 3×3 matrix with $\det A = 5$, find $\det(2A)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Determinants and scalar multiples / row operations

Swapping two rows of a matrix negates its determinant. If A has rows $\langle 1, 2\rangle, \langle 3, 4\rangle$ with $\det A = -2$, find the determinant of the matrix with rows swapped: $\langle 3, 4\rangle, \langle 1, 2\rangle$.

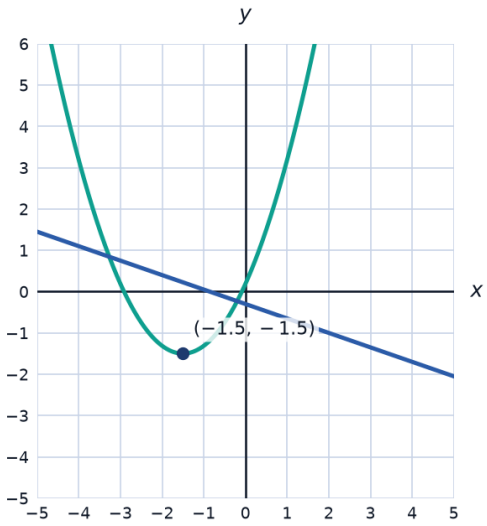
9 [Intermediate] • Target: The invertibility criterion

State the invertibility criterion in terms of the determinant.

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10 [Intermediate] • Target: The invertibility criterion

Is the matrix with rows $\langle 2, 4\rangle$, $\langle 1, 2\rangle$ invertible? Justify using the determinant.



11 [Intermediate] • Target: Applying the invertibility criterion

For what value(s) of k is the matrix with rows $\langle k, 2\rangle$, $\langle 3, k\rangle$ NOT invertible?

12 [Intermediate] • Target: Geometric meaning of the determinant

For a 2×2 matrix A with rows $\mathbf{r}_1$, $\mathbf{r}_2$, $|\det A|$ gives the area of the parallelogram spanned by $\mathbf{r}_1$ and $\mathbf{r}_2$. Find this area for $\mathbf{r}_1 = \langle 3, 0\rangle$, $\mathbf{r}_2 = \langle 1, 4\rangle$.

13 [Intermediate] • Target: Geometric meaning of the determinant

Explain, using the geometric meaning of the determinant, why a matrix with two identical rows always has determinant 0.

14 [Intermediate] • Target: Determinants of special matrices

Find the determinant of the identity matrix I_3 (rows $\langle 1, 0, 0\rangle$, $\langle 0, 1, 0\rangle$, $\langle 0, 0, 1\rangle$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

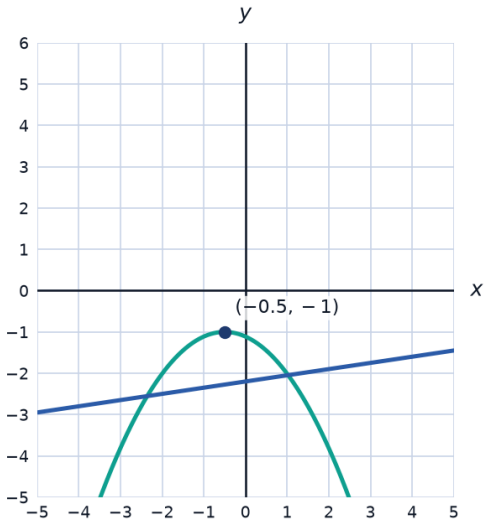
15 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Find the determinant of the matrix with rows $\langle 3, 0, 0\rangle$, $\langle 0, -2, 0\rangle$, $\langle 0, 0, 5\rangle$ (a diagonal matrix).

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16 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 1.01 Vectors in n Dimensions

Find the determinant of the matrix with rows $\langle 0, 0, 3\rangle$, $\langle 0, 1, 0\rangle$, $\langle 3, 0, 0\rangle$, and explain why it is nonzero even though the main diagonal has two zeros.



17 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 2.01 Systems of Linear Equations

Find the determinant of the zero matrix (all entries 0) of any size, and connect this to the invertibility criterion.

18 [Challenging] • Target: Determinants of special matrices • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Explain why any matrix with an entire row (or column) of zeros must have determinant 0, using cofactor expansion along that row.

19 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 1.01 Vectors in n Dimensions

Cramer's Rule solves $ax + by = e$, $cx + dy = f$ via $x = \frac{\det A_x}{\det A}$, $y = \frac{\det A_y}{\det A}$, where A is the coefficient matrix and A_x, A_y replace the respective column with $\langle e, f\rangle$. Use it to solve $2x + 3y = 8$, $x - y = 1$.

20 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 2.01 Systems of Linear Equations

Explain why Cramer's Rule fails (gives no unique solution) whenever $\det A = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: 2x2 determinants

Find the slope of the least-squares line for $(-1,\ -1)$, $(3,\ 2)$, $(4,\ 1)$.

a. $-\frac{1}{3}$

b. $\frac{3}{2}$

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

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22 [Foundational] • Target: 3x3 determinants via cofactor expansion

Given $A = [(-8, 9; -6, 7)] = PDP^{-1}$ with $P = [(3, 1; 2, 1)]$, $D = [(-2, 0; 0, 1)]$, find the (2,2) entry of A^4.

a. -29

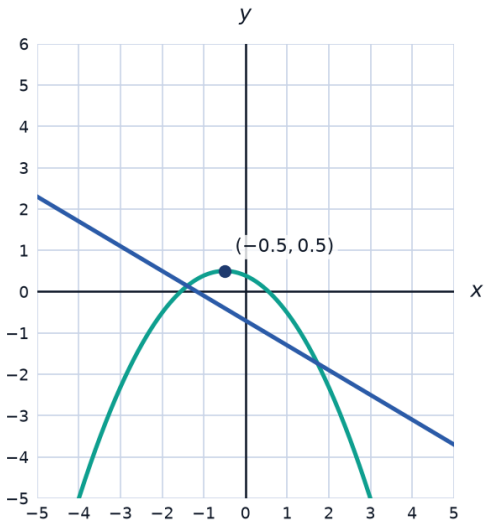
b. 2401

c. 19

d. -28

23 [Foundational] • Target: The determinant product rule

Find the distance from the point $\langle 4, -1 \rangle$ to the line through the origin spanned by $\langle 2, -2 \rangle$.



a. $\frac{5\sqrt{2}}{2}$

b. $\sqrt{17}$

c. $\frac{3\sqrt{2}}{2}$

d. $3\sqrt{2}$

24 [Foundational] • Target: Determinants and scalar multiples / row operations

Find the orthogonal projection of $\mathbf{y} = \langle -4, -2 \rangle$ onto $\mathbf{u} = \langle 0, 2 \rangle$.

a. $\langle -4, 0 \rangle$

b. $\langle \frac{4}{5}, \frac{2}{5} \rangle$

c. $\langle 0, -2 \rangle$

d. $\langle 0, -8 \rangle$

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25 [Foundational] • Target: The invertibility criterion

Which vector is an eigenvector of $A = [(3, -12; 2, -7)]$ for $\lambda = -3$?

a. $\langle -3, -1 \rangle$

b. $\langle 1, 2 \rangle$

c. $\langle 2, 1 \rangle$

d. $\langle 2, -1 \rangle$

26 [Foundational] • Target: Applying the invertibility criterion

What is the dimension of $\text{span}\{x^2 + 1, 2x^2 + 2\}$ in P_2?

a. 2

b. 1

c. 3

d. 0

27 [Foundational] • Target: Geometric meaning of the determinant

Find the orthogonal projection of $\langle -1, 4, 3 \rangle$ onto $W = \text{span}\{\langle 1, 0, 1 \rangle, \langle 1, 0, -1 \rangle\}$ (an orthogonal basis).

a. $\langle 0, 4, 0 \rangle$

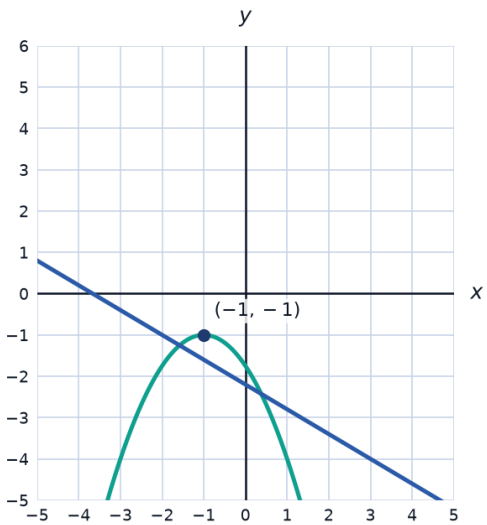
b. $\langle 1, 0, 1 \rangle$

c. $\langle -1, 0, 3 \rangle$

d. $\langle -2, 0, 6 \rangle$

28 [Foundational] • Target: Determinants of special matrices

Find the characteristic polynomial of $[(3, -4; -4, -2)]$.



a. $\lambda^2 - \lambda - 22$

b. $\lambda^2 + \lambda - 22$

c. $\lambda^2 - \lambda + 22$

d. $\lambda^2 + 22\lambda + 1$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Cramer's Rule**

$\langle 1, 1, 1\rangle, \langle 1, -1, 0\rangle, \langle 1, 1, -2\rangle$ is an orthogonal basis. Find the coefficient of $\langle 1, 1, -2\rangle$ when $\langle 5, 2, -5\rangle$ is written in this basis.

a. 17

b. $\frac{17}{3}$

c. $-\frac{17}{6}$

d. $\frac{17}{6}$

30 **[Intermediate]** • **Target: 2x2 determinants**

A has eigenvalues $2, 3$. What are the eigenvalues of $A + 2I$?

a. $2, 3$

b. $4, 6$

c. $4, 5$

d. $-4, -5$

31 **[Intermediate]** • **Target: 3x3 determinants via cofactor expansion**

Are $\langle 1, 4\rangle$ and $\langle -2, -8\rangle$ linearly independent?

a. Yes, they are independent

b. Only if one of them is the zero vector

c. It cannot be decided from the vectors alone

d. No, they are dependent

32 **[Intermediate]** • **Target: The determinant product rule**

Find the image of $\langle 3, 1\rangle$ under the reflection across the x-axis.

a. $\langle -3, 1\rangle$

b. $\langle 3, 1\rangle$

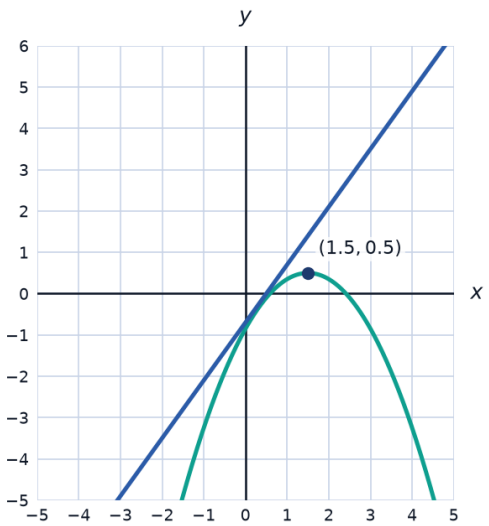
c. $\langle -1, 3\rangle$

d. $\langle 3, -1\rangle$

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33 **[Intermediate]** • **Target: Determinants and scalar multiples / row operations**

Find the image of $\langle 1, -2 \rangle$ under the scaling by the factor 3.



a. $\langle -3, 6 \rangle$

b. $\langle 1, -2 \rangle$

c. $\langle -6, 3 \rangle$

d. $\langle 3, -6 \rangle$

34 **[Intermediate]** • **Target: The invertibility criterion**

Find the image of $\langle -1, 2 \rangle$ under the orthogonal projection onto the x-axis.

a. $\langle -1, 0 \rangle$

b. $\langle 1, 0 \rangle$

c. $\langle -1, 2 \rangle$

d. $\langle 0, -1 \rangle$

35 **[Intermediate]** • **Target: Applying the invertibility criterion**

Find the image of $\langle 3, 2 \rangle$ under the horizontal shear x ■ x + 3y.

a. $\langle 3, 11 \rangle$

b. $\langle 9, 2 \rangle$

c. $\langle -9, -2 \rangle$

d. $\langle 3, 2 \rangle$

36 **[Intermediate]** • **Target: Geometric meaning of the determinant**

What is the dimension of span of $\langle 2, -3, 1 \rangle$, $\langle 2, -3, 1 \rangle$, $\langle 2, -3, 1 \rangle$, $\langle -4, 6, -2 \rangle$?

a. 4

b. 2

c. 1

d. 3

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37 [Intermediate] • Target: Determinants of special matrices

Let W be spanned by $\langle 8, -2, 4, 0, 3\rangle$, $\langle -2, 0, -2, 0, -3\rangle$, $\langle -1, 0, -2, -2, 3\rangle$, $\langle 3, -1, 1, 0, 0\rangle$ in R^5. What is $\dim W^{\perp}$?

a. 3

b. 5

c. 2

d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Cramer's Rule • Cross-Lesson: 3.01 Matrices & Matrix Algebra

Find the dimension of the null space of $[(3, 3, -2, 1, -1, -3; 0, 0, 0, 0, 0, 0; -3, -3, 2, -1, 1, 3; -3, -3, 2, -1, 1, 3)]$.

a. 1

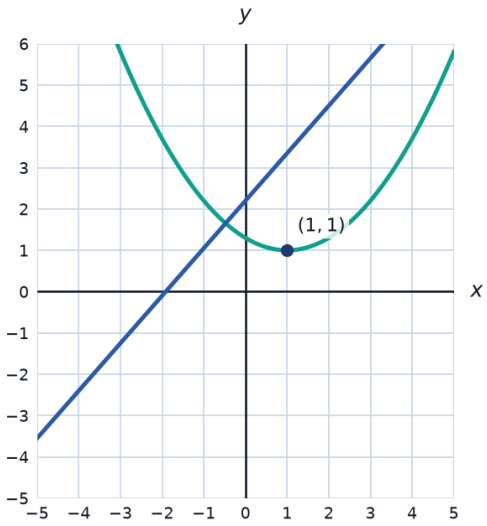
b. 5

c. 3

d. 6

39 [Challenging] • Target: 2x2 determinants • Cross-Lesson: 1.01 Vectors in n Dimensions

What is the dimension of the subspace of R^4 defined by $2x_1 - x_2 + 2x_3 - 2x_4 = 0$, $-x_1 - 3x_2 - x_4 = 0$?



a. 2

b. 4

c. 3

d. 1

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40 [Challenging] • Target: 3x3 determinants via cofactor expansion • Cross-Lesson: 2.01 Systems of Linear Equations

Find the eigenvalues of $[(4, 0; 3, 3)]$.

a. $0, 4$

b. $4, 5$

c. $-3, -4$

d. $3, 4$

41 [Challenging] • Target: The determinant product rule • Cross-Lesson: 3.01 Matrices & Matrix Algebra

$\mathbf{v} = \langle 4, 6 \rangle$ is an eigenvector of $A = [(7, -2; 6, 0)]$. What is its eigenvalue?

a. 3

b. 5

c. 4

d. -4

42 [Challenging] • Target: Determinants and scalar multiples / row operations • Cross-Lesson: 1.01 Vectors in n Dimensions

$T : R^7 R^5$ is linear with range of dimension 4. What is $\dim \ker T$?

a. 3

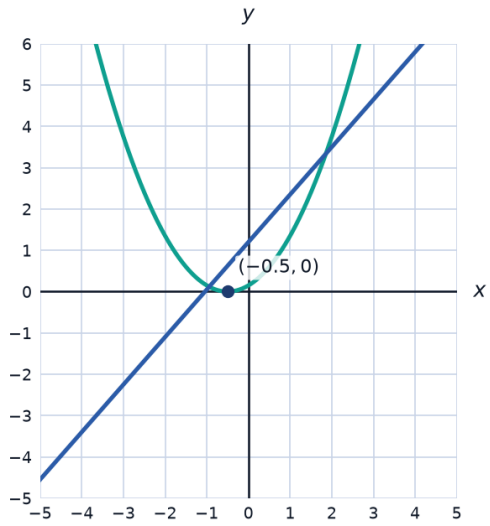
b. 1

c. 4

d. 7

43 [Challenging] • Target: The invertibility criterion • Cross-Lesson: 2.01 Systems of Linear Equations

What is the dimension of $\{A \in M_{2\times 2} : \text{tr}A = 0\}$?



a. 3

b. 4

c. 2

d. 6

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44 [Challenging] • Target: Applying the invertibility criterion • Cross-Lesson: 3.01 Matrices & Matrix Algebra

For which k are $\langle 0, -2, -2\rangle$ and $\langle -1, 2, k\rangle$ orthogonal?

a. 2

b. -2

c. -1

d. 3

45 [Challenging] • Target: Geometric meaning of the determinant • Cross-Lesson: 1.01 Vectors in n Dimensions

$T: R^6 R^4$ is linear with $\dim \ker T = 3$. What is the dimension of its range?

a. 3

b. 4

c. 1

d. 6

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